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Choline hydroxide based deep eutectic solvent for dissolving cellulose†

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New cellulose solvent based on choline hydroxide/urea (ChOH/Ur) deep eutectic solvent (DES) was developed. The ChOH/Ur DES has a promising solubility for cellulose compared to ChCl-based DESs. ChOH/Ur DES with *ca.* 11 wt% water can dissolve at least 9.5 wt% cellulose with the degree of polymerization (DP) of 926 within 30 min. Compared to the ionic liquids, the temperature window of ChOH/Ur is wide and close to room temperature. The ChOH/Ur DES also showed more tolerance to water than ionic liquid, which can save energy in preparation and usage. The physical and chemical properties of DES and cellulose solution were investigated. The ChOH/Ur DES system is a non-derivatizing solvent for cellulose. The preliminary rheological results suggest that the dynamics of cellulose solution in ChOH/Ur system is similar to that of salt-free polyelectrolyte solutions. The degradation of cellulose during the dissolution and storage of the cellulose solution is limited. After 7 days of storage at 40 °C, the DP of the regenerated cellulose remained at about 80% of the raw material. The ChOH/Ur is recyclable without the loss of solubility of cellulose. In conclusion, the ChOH/Ur DES system is a promising green solvent platform for preparing regenerated cellulose materials and could be used as a media for cellulose chemistry.

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Introduction

Cellulose, composed of linear β-(1→4)-linked glucose repeating units, is one of the most abundant natural biopolymers on the Earth and produced up to a thousand billion tons each year *via* photosynthesis. With the increasing demand for environmentally friendly and biocompatible products,[1–5] cellulose is considered as an inexhaustible source and a promising candidate for bio-based chemicals and fuels.[6–10] However, the chain stereoregularity and abundant hydroxyl groups in cellulose lead to the formation of a strong hydrogen bonding network *via* the inter- and intra-chain hydrogen bonds. Thus, cellulose cannot be melted or dissolved in common solvents, which limits the application of cellulose, and only about 0.2% cellulose is utilized, mostly in the form of paper, textile, *etc*. Thus, finding solvents to break the hydrogen bonds of cellulose is crucial for the production of regenerated cellulose materials and the modification of cellulose into functional materials.

Cellulose solvents have been developed for more than a century. By now, it can be classified into derivatizing and direct (non-derivatizing) solvent systems. The viscose process for producing regenerated cellulose fibers and films is a classical derivative system, which is still prevalent. In the viscose process, cellulose reacts with CS_2 to form cellulose xanthonate and is then dissolved in NaOH aqueous solution to produce regenerated cellulose materials.[11] The use of CS_2 and release of toxic by-products in the viscose process cause serious environmental pollution and viscose plants have been gradually closed since 1980s in developed countries. Therefore, researchers are devoted to finding direct solvents of cellulose. The represented systems include *N*,*N*-dimethylacetamide/lithium chloride (DMAc/LiCl),[12–14] *N*-methylmorpholine-*N*-oxide (NMMO) mono-hydrate,[15] ionic liquids (ILs),[16,17] and NaOH/urea aqueous solvents.[18] However, these cellulose solvents still possess many problems. For example, pre-treatments *via* solvent exchange or high temperature are needed before dissolving cellulose in DMAc/LiC.[19] Imidazole-based ionic liquid has an outstanding solubility for cellulose but is expensive with high energy consumption.[20] Alkali urea aqueous system needs to previously freeze the solvent at a temperature below −12 °C and can only dissolve cellulose with relatively low molecular weight (*ca.* DP < 600).[21] Thus, as a low-cost green solvent system, deep eutectic solvents (DESs) have attracted increasing attention recently.

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DES is a mixture of hydrogen bond acceptor (HBA, *e.g.*, quaternary ammonium salts) and hydrogen bond donor (HBD, *e.g.*, amines, amides, carboxylic acids, and polyols).[22–25] The earliest DES was reported by Abbott, a mixture of choline chloride and urea.[22] The melting point of DESs is much lower than either of the individual components, which is due to the increase in the entropy of the system *via* the hydrogen bonding association of HBD and HBA.[26] In addition, the physical and chemical properties of DESs can be readily regulated owing to the numerous ($>10^6$) possible combinations between HBA and HBD. DESs also have the advantages of easy preparation, low vapour pressure, non-flammability, biocompatibility, and biodegradability, depending on the species that used.[23,27,28] DES is a promising solvent system for various solutes and chemical reactions.[25,29,30] Because of the hydrogen bonding interaction of HBD and HBA, DESs could be a potential solvent system to dissolve cellulose as ILs. Several works reported the dissolution of cellulose in DESs.[31–38] However, these DESs show poor solubility of cellulose and are invaluable for producing regenerated cellulose materials. Therefore, recent works of DESs mainly focused on pre-treatment, separation, and conversion of biomass.[39–42] Inspiringly, recent reports indicate that $ZnCl_2$ based DESs have a sound solubility of cellulose and can be used to prepare regenerated cellulose materials under mild conditions.[43,44]

In this work, our purpose is to find an efficient, economical, and environmentally friendly solvent for cellulose. After systematical research, we discovered a choline hydroxide/urea (ChOH/Ur) DES system as a sound direct solvent for cellulose. The degradation of cellulose during dissolution and regeneration procedures is limited. The ChOH/Ur DES is prepared from widely used chemicals and can be used near room temperature, from 30–70 °C, which offers to save the system cost and energy. ChOH is primarily used to produce choline salts, a kind of food nutrient additive and safe enough to be used in baby food. ChOH itself is also widely used as the cleaning and degreasing agent in the metal industry. Urea is a biological metabolite and used as fertilizer. ChOH/Ur shows the advantages of high water tolerance and excellent recyclability. Our results suggest that ChOH/Ur system is a promising cheap solvent for cellulose, which could be developed into a state-of-the-art environmentally friendly platform for preparing regenerated cellulose materials and a medium for cellulose chemistry.

Experimental

Materials

Choline chloride (ChCl; 99%, Adamas-Beta) was recrystallized from absolute ethanol and vacuum dried at 50 °C overnight before use. The concentration of choline hydroxide (ChOH, ~45 wt%, Acros) aqueous solution was determined by titration with 0.1 M HCl solution. Urea (99%), thiourea (99%), resorcinol (99%), acetic acid (99%), L-histidine (99%), serine (99%), KOH (≥85%), and imidazole (99%), supplied by local chemical suppliers, are analytical grade and used as received. Microcrystalline cellulose (MCC, Alfa Aesar) and cotton linter pulp (Shandong Yinying Co., Ltd, China) were vacuum dried overnight before use. The degree of polymerization (DP) of MCC (DP = 145) and cotton linter pulp (DP = 541 and DP = 926) was measured by intrinsic viscosity measurements in 0.5 M cupriethylenediamine (CED) at 25 °C.

Preparation of DESs

The ChCl-based DESs were prepared according to the literature.[30,38,45,46] Briefly, ChCl was mixed with HBDs in a specific ratio and then heated to 60–70 °C with stirring until a clear homogenous liquid was formed. In the preparation of ChOH-based DESs, ChOH aqueous solution with known concentration was condensed by vacuum rotary evaporation at 67 °C, during which the vacuum gradually reduced from 1 kPa to 0.2 kPa. The relatively mild conditions could prevent the decomposition of ChOH during the condensing procedure. Water content was calculated from the loss of mass. The condensed ChOH was then mixed with HBDs to form DESs, following the same procedure as above. The details of the preparation conditions are listed in Table 1.

Table 1 The details for the preparation of DESs and their solubility of MCC

HBA	HBD	Molar ratio	Dissolving temp. (°C)	Dissolution time	Solubility (wt%)
ChCl	Urea	1 : 2	100	24 h	Non
	Thiourea	1 : 2	100	24 h	Non
	Resorcinol	1 : 1	100	24 h	Non
	Imidazole	3 : 7	100	24 h	Non
	Acetic acid	1 : 1	60	24 h	Non
	KOH[a]	1 : 4	100	—	—
	L-Histidine[b]	1 : 1	100–140	—	—
ChOH[c]	Urea	1 : 2	70	20 min	9.51
	Thiourea	1 : 2	70	2 h	Non
	Resorcinol	1 : 1	100	2 h	Non
	Imidazole	3 : 7	100	2 h	Non
	L-Histidine	1 : 1	100	2 h	Non
	Serine	1 : 1	100	2 h	Non

[a] This combination cannot form DES without water. [b] This combination cannot form DES from 100 °C to 140 °C. [c] With 20 wt% water.

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Solubility measurements and regeneration of cellulose

The solubility of cellulose in DESs was checked using the high crystallinity of MCC. ChCl-based DESs need enough time to dissolve cellulose. Thus, the freshly prepared ChCl-based DESs were divided into several equal parts, and a certain amount of MCC was added. The content of MCC in each part of DESs cellulose was in the range of 0–3 wt% with an interval of 0.5 wt%. The mixtures were heated under stirring at 100 °C for 24 h, followed by solubility measurements. The solubility of cellulose in DESs was estimated as follows. The above-prepared cellulose/DES mixtures were sandwiched between glass slides and observed under a polarized microscope (POM, Olympus EP50). A dark field in the view indicates the complete dissolution of MCC. Each experiment was repeated twice and the average was taken as the result.

For ChOH-based DESs, MCC was added in batches, and solubility estimation was carried out in 30 min after each addition of MCC. If completely dissolved, continued adding MCC until the insoluble matter was observed and recorded the concentration. Regenerated cellulose samples were prepared by pouring the cellulose solution into a coagulation bath of water or ethanol to precipitate cellulose and collected by filtration. Then, the regenerated cellulose samples were washed thoroughly with deionized water and vacuum dried at 70 °C overnight.

Kamlet–Taft parameters measurement of DESs

The Kamlet–Taft (KT) parameters of DESs were measured using the solvatochromic dyes of 4-nitroaniline (4NA), *N*,*N*-diethyl-4-nitroaniline (DENA), and Reichardt's dye (RD). In detail, 2 mg dyes were separately dissolved in 4 mL ethanol. A certain amount of the dye solution was transferred into a vial, and the ethanol was evaporated at 40 °C for 12 h. Then, DESs (2.5 g) was added to the vial and stirred to dissolve the dye completely. UV-vis spectra in the range of 350–700 nm of the dyed DESs were measured on a UV-vis spectrophotometer (UH4150, Hitachi) at 25 °C in a quartz cell of 1.0 mm light-path length. The maximum absorbance of the UV-vis spectra was controlled in the range of 0.4–2 by adjusting the volume of the dye solution added in the vials. The KT parameters, say the dipolarity–polarizability (π^*), hydrogen bond acidity (α), and hydrogen bond basicity (β) were calculated by,[47–49]

$$\pi^* = \frac{\nu_{\max,\mathrm{DENA}} - 27.52}{-3.182} \quad (1)$$

$$\alpha = \frac{E_T(30) - 14.6(\pi^* - 0.23) - 30.31}{16.5} \quad (2)$$

$$\beta = \frac{1.035\nu_{\max,\mathrm{DENA}} - \nu_{\max,4\mathrm{NA}} + 2.64}{2.8} \quad (3)$$

where $\nu_{\max}$ (cm^{-1}) is the frequency at the wavelength of maximum absorbance $\lambda_{\max}$ (nm) of the corresponding dyes and $\nu_{\max} = 10^4/\lambda_{\max}$. $E_T(30)$ (kcal mol^{-1}) expresses the solvent polarity arising from the overall interactions between a solvent and the dye and $E_T(30) = 28\,591/\lambda_{\max,\mathrm{RD}}$.

Measurements and instruments

The thermal properties of DESs were investigated using differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA). DSC measurements were carried out on a differential scanning calorimeter (DSC Q2000, TA Instruments, USA). The DESs samples were first equilibrated at 20 °C. Then, a cooling–heating cycle program between 20 °C and −80 °C was run with a ramp of 10 °C min^{-1}. TGA measurements were performed on a thermogravimetric analyzer (Pyris 1 TGA, PerkinElmer, USA). The DESs samples were heated from room temperature to 400 °C at a heating rate of 10 °C min^{-1} in a nitrogen atmosphere.

Fourier transform infrared (FTIR) spectra were recorded on a Bruker Tensor27 FTIR spectroscopy. Each spectrum was recorded in the range of 4000–400 cm^{-1} at the resolution of 4 cm^{-1} with the accumulation of 32 scans. KBr pellets were used for sample preparation.

Wide angle X-ray diffraction (WAXD) measurements were carried out on a PANalytical Empyrean diffractometer with the $\mathrm{Cu}(\mathrm{K}_\alpha)$ radiation source (λ = 1.54 Å). The diffractometer was operated at 40 kV and 40 mA at the scanning rate of 2° min^{-1} in the range of 2θ of 5°–40°.

Rheological properties of cellulose solutions were measured using ARES-G2 Rheometer (TA Instrument, USA). Two cone-plate geometry sets, 20 mm of 1° angle cone and 40 mm of 2° angle cone, were used according to the concentration of the cellulose solutions. Steady shear and dynamic oscillatory shear experiments were carried out in the shear rate range of 0.01–100 s^{-1} and frequency range of 0.1–100 rad s^{-1}, respectively. For the measurement in oscillation mode, a strain sweep from 0.1 to 30% with a fixed frequency of 1 rad s^{-1} was performed to determine the linear viscoelastic regime before each frequency sweep. A strain of 5% was chosen to fall well into the linear viscoelastic regime for the frequency range of 0.1–100 rad s^{-1} in the oscillatory shear measurements. The experimental temperature was set at 25 °C, 35 °C, 45 °C, and 55 °C. A thin layer of low-viscosity silicone oil was smeared at the edge surface of the sample to avoid the influence of water evaporation or absorption on the rheological properties of the solutions during measurement.

Stability of cellulose in ChOH/Ur solution

The stability of cellulose in ChOH/Ur solution was estimated as follows. The degree of polymerization (DP) of cellulose was determined using Ubbelohde viscometer. Cotton linter pulp was dissolved in ChOH/Ur, stored for different days at 70 °C, and then regenerated with deionized water. The samples were vacuum dried before DP measurement. To eliminate the mechanical effect on DP, such as agitation and shear, the raw pulp was treated with the same procedure as those regenerated cellulose samples. The DP of cellulose samples was evaluated from intrinsic viscosity measurements in 0.5 M cupriethylenediamine (CED) at 25 °C using the relationship reported by Marx.[50]

$$[\eta]\ (\mathrm{mL\ g}^{-1}) = 0.98\mathrm{DP}^{0.9} \quad (4)$$

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Results and discussion

Solubility of cellulose in DESs

The solubility of cellulose in DESs prepared from different pairs of HBAs and HBDs reported in the literature was first checked. The results are listed in Table 1. Unlike the relatively high solubility of cellulose in DESs reported in the literature, *e.g.*, 8 wt% in ChCl/Ur,[46] 10 wt% in ChCl/Thio,[46] 6.1 wt% in ChCl/Res,[45] and 2.48 wt% in ChCl/Im,[38] we found that almost all the DESs show negligible solubility of cellulose. In our experiments, crowding crystalline particles were clearly observed under a polarizing microscope even when the concentration of MCC was less than 0.5 wt% (Table S1, ESI†). The deviation from the literature could be attributed to the raw materials and the measuring method. In our measurements, no pre-treatment was performed on the raw materials, such as boiling or mercerizing, which could reduce the degree of crystallinity of MCC, and no third component was added into the ChCl-based DESs, such as water. It was reported that DES composed of ChCl and KOH can dissolve cellulose.[51] However, we found that no DES can be formed and cellulose cannot be dissolved in this mixture. Based on previous works[14,52,53] and the results in the literature,[27] the higher basicity the HBA has, the stronger the ability to weaken the inter- and intra-molecular hydrogen bonds of cellulose. Therefore, the chloride ion in choline chloride was substituted by the hydroxide ion. It should be noted that it is very difficult to completely remove water in ChOH aqueous solutions. Thus, ChOH aqueous solution was only condensed to 80 wt%. Different HBDs were added in it to form ChOH based DESs. We found that only ChOH/Ur system has a relatively higher solubility of cellulose among the ChOH based DESs prepared in the present work (Table 1). By comparing the components of the DES pairs of ChCl/Ur, ChOH/Ur, and ChOH/thiourea, it can be speculated that the hydroxide ion and the carbonyl groups of urea should be critical for breaking the hydrogen bond network of cellulose and stabilizing cellulose chains in the media to form a homogeneous cellulose solution.

The most attractive fact is that ChOH/Ur system is insensitive to water for dissolving cellulose as that of NMMO,[54] ionic liquids,[55,56] and DMAc/LiCl.[57,58] Fig. 1 shows the solubility of MCC in ChOH/Ur as a function of water content. The inseted photo shows the typical homogeneous solution that was prepared. POM observation indicates that MCC and pulp can be dissolved within 30 min (Fig. S1–S3, ESI†). Vigorous stirring of the mixture promotes the dissolution procedure significantly. The solubility of MCC in ChOH/Ur aqueous system decreases with the increase in the water content in the system (Fig. 1). It is reasonable that water molecules participate in the hydrogen bond formation inside the system.[59] The interaction of water molecules and solvent system competing with that of cellulose molecules and solvent system results in reducing the solubility of cellulose. Related theoretical investigation indicates that the presence of water accelerates the hydrogen-bond dynamics and abates heterogeneity in DES.[60] But the influence of water on the solubility of cellulose in ChOH/Ur system is much lower

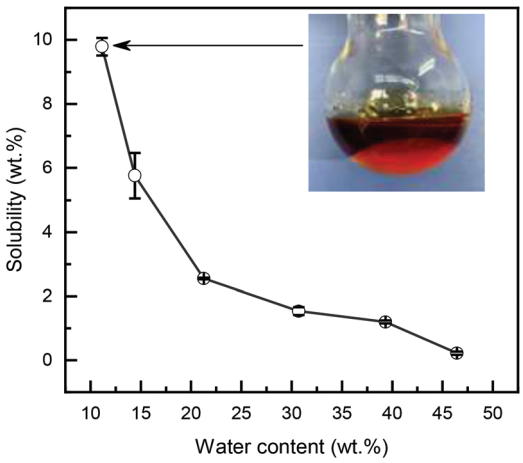


Fig. 1 The solubility of MCC in ChOH/Ur (1 : 2) as a function of water content at70 °C. The inset picture shows the resultant cellulose solution.

than that in ionic liquid, in which cellulose is insoluble with the water content >1.0 wt%.[16] In this work, the ChOH/Ur DES still has good solubility for cellulose with the water content of 10–15 wt%. This means that energy consumption to remove water can be reduced during the preparation and recycling procedure of the solvent system. It should be noted that the solubility of MCC in ChOH/Ur aqueous system in this work was estimated at laboratory level using a magnetic bar for stirring, and the solution at a relatively higher cellulose concentration is too viscose to be stirred by the magnetic stirring bar. In principle, the dissolving process of a polymer is thermodynamically and kinetically controlled. It is sure that the solubility and dissolving process can be promoted by using a more effective mixing method. Cellulose solution with even higher cellulose concentration is possible to be prepared. Higher concentration is important to improve both the processing efficiency and the physical properties of the regenerated cellulose materials.

Fig. 2 shows the solubility of MCC in ChOH/Ur system (water content ~11.8 wt%) as a function of temperature. The solubility of cellulose in ChOH/Ur aqueous system is obviously temperature dependent. The solubility of cellulose in ChOH/Ur aqueous system first increases with the rising temperature and reaches the maximum value at 70 °C, at which about 10 wt% MCC can be dissolved. Further increase in temperature leads to the decrease in the solubility of cellulose. In the temperature range of 30–80 °C, the solvent system shows a sound solubility of cellulose, which suggests that the solvent system has a wide window for dissolving cellulose. In fact, 40–50 °C is satisfactory for dissolving cellulose and is close to room temperature. We observed the formation of bubbles in cellulose/ChOH/Ur with the release of ammonia smelling gas at the temperature above 90 °C during cellulose solubility determination. To investigate possible degradation in the system, the cellulose/ChOH/Ur mixture was kept at 90 °C for 5 h and a new strong absorbance band at 2154 cm^{-1} appears in the FTIR spectrum (Fig. S4, ESI†), assigned to [NCO^-] groups.[61] The

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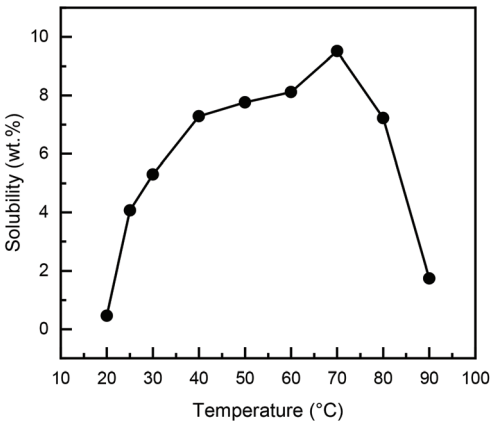


Fig. 2 Solubility of MCC in ChOH/Ur (1 : 2) aqueous system as a function of temperature. Water content in the ChOH/Ur aqueous system is 11.8 wt%.

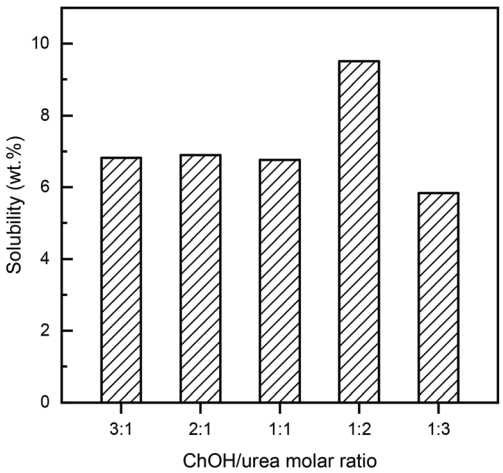


Fig. 3 Solubility of cellulose in the ChOH/Ur solvent with different ChOH/Ur ratios. Water content in ChOH for preparing the solvents is 20 wt%.

results confirm that the decrease in solubility of cellulose in the ChOH/Ur at high temperatures (*e.g.*, above 90 °C) is attributed to the degradation of urea in the system. It is noted that the dissolving temperature of cellulose in ILs is in the range of 80–110 °C, *e.g.*, 80 °C in 1-allyl-3-methylimidazolium chloride and 90 °C in 1-ethyl-3-methylimidazolium acetate.[62] The temperature for dissolving cellulose in NMMO·H_2O is above 80 °C. Moreover, NMMO is also a strong oxidant and is rather sensitive towards quite a number of chemical species, such as alkylating and acylating agents as well as redox-active transition metal ions. The plethora of complex homolytic (radical) and heterolytic (ionic) degradation reactions in the NMMO·H_2O/cellulose system cause uncontrollable exothermic reactions with explosions, bursts of flames, and destruction of the equipment and building structure.[63] Therefore, the usage conditions of the ChOH/Ur system for dissolving cellulose are quite mild compared to those of ILs and NMMO monohydrates. Besides MCC, the solubility of cellulose raw materials with a high DP, cotton linter pulp (DP = 926), was also checked. The solubility of the cotton linter pulp in ChOH/Ur aqueous system with a water content of 11.8 wt% is 9.63 wt% at 70 °C, which is similar to the solubility of MCC. In principle, the solubility of a polymer in solvents is thermodynamically determined, which is irrelevant to the molar mass. The dissolving rate of cellulose with high DP becomes slower since it is kinetically unfavourable in the dissolution of polymers with higher molar mass. The little difference in the solubility of cellulose samples with different DP should be attributed to the experimental errors. The ChOH/Ur aqueous solvent system has a higher solubility than that of low temperature NaOH/urea aqueous system, which has a solubility limit of cellulose with relative low DP.[18] The promising solubility for cellulose with high molecular weight is very important for developing the process to produce regenerated cellulose materials with desired properties.

The dependence of the solubility of cellulose on ChOH/Ur molar ratio in the solvent system was investigated, and the results are shown in Fig. 3. In all the experiments, the water content in the ChOH/Ur aqueous system is fixed at 20 wt% to ChOH. The results indicate that the solubility of cellulose in the solvent system is relatively high (>6.0 wt%) in all ChOH/Ur molar ratios investigated. The maximum value of the solubility of cellulose (~9.5 wt%) was achieved at the ChOH/Ur molar ratio of 1 : 2. The results show this solvent system does not need the strict requirement for controlling the water content and ChOH/Ur molar ratio. The main difference for the solvent system with different water contents and ChOH/Ur molar ratio is the difference in the viscosity and freezing point.

Kamlet–Taft parameters and solubility of cellulose

Solvatochromic dyes can be used as solvent polarity indicators, by which KT parameters can be determined, including the dipolarity–polarizability (π^*), hydrogen bond acidity (α), and hydrogen bond basicity (β) of the solvent.[64] The KT parameters of the DESs investigated in this work are listed in Table 2. The data of $E_T(30)$ and α for ChCl/Res and ChOH/Res are absent since no absorption peak was found in the RD solutions. ChCl/Im, ChCl/Thio, and ChOH/Thio are solids at 25 °C, and their KT parameters were not measured. The KT parameters of DMSO and ChCl/Ur measured in this work are similar to those

Table 2 The Kamlet–Taft parameters of DESs, where the molar ratios of the solvents are the same as Table 1[a]

Solvent	$E_T(30)$	π^*	α	β
ChCl/Ur	57.95	1.20	0.82	0.49
ChCl/Res	—	1.14	—	0.31
ChOH/Ur	58.08	1.32	0.72	1.88
ChOH/Ur–30% H_2O	62.14	1.30	0.98	0.46
ChOH/Res	—	1.15	—	0.53
ChOH/Im	47.65	1.18	0.21	0.57
DMSO	44.9	1.01	0.19	0.73

[a] All the measurements were carried out at 25 °C.

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in the literature.[65,66] The β value of the solvent is generally used to evaluate the hydrogen bond acceptor ability, which is highly correlated with cellulose solubility.[67] Some solvents, such as ILs, with $\beta > 0.8$ can easily form hydrogen bonds with hydroxyl groups of cellulose and has good cellulose solubility.[68] The β value of ChOH/Ur is 1.88, which is much higher than the β value of other DESs (<0.6) (Table 2). The high hydrogen bond basicity offers an excellent solubility of cellulose in ChOH/Ur DES, while the other DESs having a low β value (<0.6) and weak hydrogen bond basicity cannot dissolve cellulose.

Comparing the KT parameters of ChCl/Ur and ChOH/Ur, the α values are close to each other, but the β values are quite different. It can be speculated that the anions of the DES HBA have less effect on the hydrogen acidity but have a strong effect on the hydrogen basicity of DES. Therefore, replacing the anion of the choline based DESs with high hydrogen bond acceptor ability is expected to improve the solubility of cellulose. It can also indicate that hydroxide ion has stronger hydrogen bond acceptor ability than chloride ion, which is in line with reality. To clarify the influence of the water content in ChOH/Ur on the solubility of cellulose, the KT parameters of ChOH/Ur with 30 wt% water (ChOH/Ur–30% H_2O) were measured. The β value of ChOH/Ur–30% H_2O dropped to 0.46, and the α value increased to 0.98 at 25 °C (Table 2). It is known that water has a low hydrogen bond basicity (β = 0.18) and a high hydrogen bond acidity (α = 1.17).[69,70] The changes in KT parameters of ChOH/Ur with the addition of water in the system could be attributed to the strong interaction between water molecules and the OH^- ions of ChOH/Ur and weaken the hydrogen bond basicity of the solvent. The presence of excess water molecules can compete with cellulose in the formation of hydrogen bonds, which leads to reducing the solubility of cellulose in ChOH/Ur DES. Molecular dynamic simulation and vibrational analysis of ChCl/Ur (1 : 2) indicated that there is a strong interaction between $-NH_2$ of urea and chlorine anion where the system tends to maximize the number of hydrogen bonds to the anion,[71,72] which could reduce the ability for the chlorine anion to break the hydrogen bonds of cellulose.

The KT parameters of CHOH-based DESs indicate that the HBD species also have a significant impact on the β value. ChOH/Res and ChOH/Im have the β value of 0.53 and 0.57, respectively (Table 2). The results suggest that HBD selection is another important factor to dissolve cellulose. That is different from pure ILs, of which β is dominated by anionic species.[73] In the case of DESs, β is influenced by both the HBA anion and the HBD component, which could be attributed to the interactions between HBA and HBD in the formation of DESs. The formation of DESs *via* the establishment of hydrogen bonds between HBD and HBA results in a decrease in the capacity to engage in further hydrogen bond networks.[74] In other words, strong hydrogen bonds between HBA and HBD will lead to a decrease in β value, which is unfavourable for dissolving cellulose. That means cellulose and the HBD of DESs are somewhat in a competitive relationship. If the hydrogen bonding interaction between HBD and HBA of DES is stronger than that between cellulose and HBA, the β value of the DESs and their solubility of cellulose would be lowered as the ChOH-based DESs other than ChOH/Ur in this work (Tables 1 and 2). In principle, a low α value and a high β value of the HBD itself is preferred in preparing DESs to dissolve cellulose, which seems to contradict the identity as "hydrogen bond donor". According to this speculation, DMSO, with α = 0.19, β = 0.73, could be a good candidate for HBD in preparing DES for dissolving cellulose. We found that the ChOH/DMSO (1 : 2) system could dissolve MCC at least 10 wt%. The details of the ChOH/DMSO system will be reported in our future work. Urea has been reported to promote the solubility of cellulose in alkali[18,53] and TBAH[75] aqueous solvent systems, in which urea helped to stabilize the isolated cellulose chains in the solution. In ChOH/Ur system, the ChOH and urea act as HBA and HBD, respectively, to form a DES *via* strong hydrogen bonding interactions. As a new DES, the structure of ChOH/Ur DES is still not fully understood. Speculated from the results of ChCl/Ur system,[71,72] there is a strong interaction between $-NH_2$ of urea and anion of chlorine hydroxide, which forms moderate hydrogen bonds in the ChOH/Ur DES system. Therefore, the function of urea in ChOh/Ur system is different from that in the alkali and TBAH systems. More experimental and theoretical works are needed to clarify the mechanism of the dissolution of cellulose in ChOH/Ur system.

Rheology of cellulose/ChOH/Ur solutions

Steady and complex viscosity of cellulose/ChOH/Ur aqueous solutions with different cellulose concentrations at different temperatures were measured. The DP of the cellulose is 541 for all the rheological measurements. Fig. 4a shows the steady shear solution viscosity as a function of shear rate. The results indicate that the pure ChOH/Ur DES solvent shows a Newtonian behaviour. The cellulose/ChOH/Ur solutions show the typical shear-thinning behaviour at higher shear rates, and the onset of shear thinning shifts to a lower shear rate with the increase in cellulose concentration. The results are similar to those of cellulose/ionic liquid solutions.[76,77]

However, the complex viscosity value obtained by the dynamic rheological curve is not equal to the steady-state apparent viscosity value, indicating that the cellulose/ChOH/Ur DES solution system does not conform to the Cox–Merz rule of $|\eta^*|(\omega) = \eta(\dot{\gamma})|_{\omega=\dot{\gamma}}$. Our results indicate that the complex viscosity (η^*) is generally lower than the steady shear viscosity (η). A similar phenomenon has been found in cellulose/[BMIM]Cl solution[78] and dilute cellulose/NaOH/thiourea solution,[79] which could be attributed to the large persistence length of cellulose or might be due to the strong association interactions, such as hydrogen bonds and two-phase system of the polymer micro-aggregates dispersed system.[80] The steady-state flow curve of these systems can be fitted by Carreau model to estimate the viscosity of the solution at zero shear rate (η_0),[81]

$$\eta = \eta_\infty + \frac{\eta_0 - \eta_\infty}{[1 + (\tau\dot{\gamma})^2]^{(1-n)/2}} \quad (5)$$

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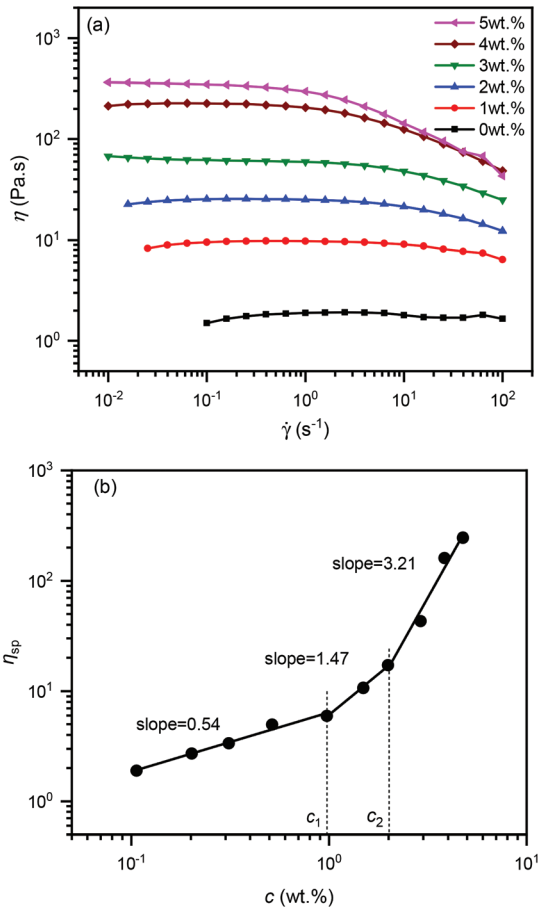


Fig. 4 (a) Steady shear viscosity (η) as a function of shear rate ($\dot{\gamma}$) and (b) the concentration dependence of specific viscosity of the cellulose (DP = 541) solutions in the ChOH/Ur (1 : 2) system at 25 °C. Water content in ChOH/Ur (1 : 2) aqueous system for preparing cellulose solutions is 11.8 wt%.

where η_∞ is the solution viscosity at the shear rate of infinite and η_∞ = 0 for polymer solutions. To eliminate the influence of DES solvent viscosity on the viscosity of the cellulose solution, the specific viscosity (η_{sp}) of the cellulose/ChOH/Ur solution can be calculated by $=(\eta_0 - \eta_s)/\eta_s$, where η_s is the solvent viscosity.[77] Polymer solution can be divided into dilute, semi-dilute, and concentrated regimes according to scaling theory.[82] In the dilute regime, the polymer chains isolated from each other start to contact each other at the overlap concentration (c^*). In the semi-dilute regime ($c^* < c < c_e$), polymer chains overlap each other without entanglement. Further increase in the polymer concentration at entanglement concentration (c_e), the overlapped polymer chains start to form a roughly uniform entangled network of various segments.

Fig. 4b shows η_{sp} of the cellulose/ChOH/Ur solution as a function of concentration. The concentration dependence of specific viscosity for solutions of cellulose (DP = 541) in ChOH/Ur shows three different power law regimes: $\eta_{sp} \sim c^{0.54}$ below c_1 = 1.0 wt%, $\eta_{sp} \sim c^{1.47}$ in the range of $c_1 < c < c_2$ (=2.0 wt%), and $\eta_{sp} \sim c^{3.21}$ above c_2. Scaling theory predicts the concentration-dependent indices of specific viscosity of 1, 2, and 14/3 in θ solvent, and 1, 1.3, and 3.9 in a good solvent for neutral polymers in dilute, semi-dilute, and concentrated regimes, respectively, which have been reported in cellulose solutions in ILs.[76,77] However, the viscosity data in this work follows the power law of salt-free polyelectrolyte solutions, which is expected of $\eta_{sp} \sim c^{1/2}$ in semi-dilute unentangled regime, $\eta_{sp} \sim c^{3/2}$ in the semi-dilute entangled regime, and $\eta_{sp} \sim c^{15/4}$ in the concentrated regime.[83] The results may be attributed to the formation of a polyelectrolyte-like complex during the dissolution of cellulose in the ChOH/Ur DES system. The details of the dynamics of cellulose chains in the ChOH/Ur DES system will be investigated in future work.

The viscosity–temperature dependence of cellulose solution in ChOH/Ur DES solution can be described by Arrhenius equation $\ln \eta_0 = E_a/RT + \ln A$, where E_a is the viscous flow activation energy, R is the universal gas constant, and T is the absolute temperature. E_a can be calculated by the linear fit of $\ln \eta_0$–$1/T$, as shown in Fig. 5. The calculation results indicate that E_a increases monotonically with increasing concentration, which is 64.2, 67.8, and 69.5 kJ mol^{-1} for cellulose solution in the ChOH/Ur system with the cellulose concentration of 3.0, 4.0, and 5.0 wt%, respectively. The activation energy is in the same order of magnitude as those of cellulose solutions in ionic liquids, such as 49 kJ mol^{-1} for 3 wt% cellulose (DP = 1000) solution in [EMIM]Ac and 57 kJ mol^{-1} for 3% cellulose (DP = 950) solution in NMMO.[84] However, the relatively higher viscous flow activation energy of the cellulose (DP = 541) solution in the ChOH/Ur system suggests that the viscosity of cellulose solutions in this work is more sensitive to temperature.

For the purpose of investigating the viscoelasticity of cellulose solutions in the ChOH/Ur system in a wide possible frequency range, we superimposed the dynamic modulus curves obtained at different temperatures through the time–temperature superposition (TTS) principle to form a master curve (Fig. 6). The curves of G' and G'' with ω at different tempera-

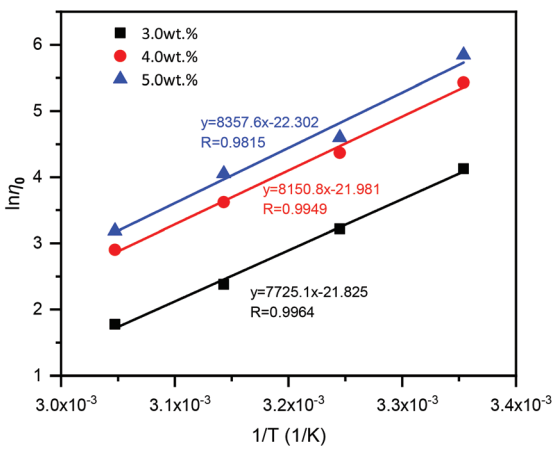


Fig. 5 The plot and linear fitting of $\ln \eta_0$–$1/T$ of cellulose (DP = 541) solutions in the ChOH/Ur (1 : 2) system. Water content in ChOH/Ur (1 : 2) aqueous system for preparing cellulose solutions is 11.8 wt%.

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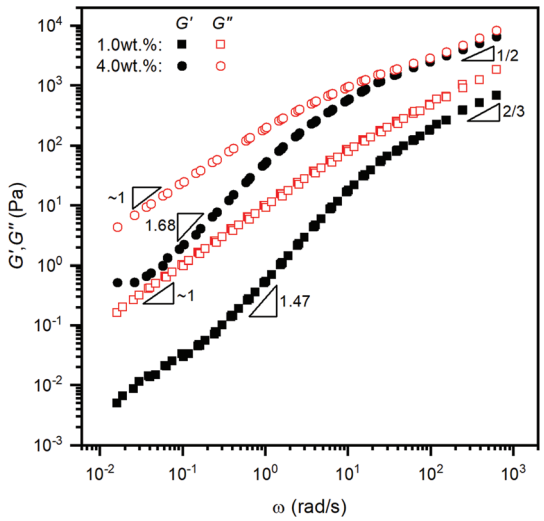


Fig. 6 Master curves of cellulose (DP = 541) solutions in the ChOH/Ur (1 : 2) system at 25 °C. Water content in ChOH/Ur (1 : 2) aqueous system for preparing cellulose solutions is 11.8 wt%.

tures can be superimposed together, indicating that the TTS principle works well within the temperature and concentration range. In the entire frequency range, the loss modulus (G'') is higher than the storage modulus (G'). The results indicate that the solution is in a viscous fluid state. At low values of ω, often referred to as the "terminal region", the slope of logG' versus logω of the cellulose solutions plots is around 1.5, while the slope of logG'' versus logω plots is 1, which is different from both Rouse and Zimm theories with the power law of $G' \sim \omega^2$ *and* $G'' \sim \omega$. This upward trend with the decrease in ω at the low values of ω may be the so-called "second plateau", which is generally related to the special viscoelastic response of multiphase polymer systems.[85] The results of dynamic oscillatory shear experiments seem to indicate the formation of a multiphase in cellulose solution even though it looks macroscopically homogeneous.

Structure of regenerated cellulose from ChOH/Ur

Fig. 7 shows the FTIR spectra of MCC and regenerated cellulose sample from ChOH/Ur system. The absorbance bands at 3340 and 3405 cm^{-1} of MCC come from the stretching mode of the inter- and intra-molecular hydrogen-bonded –OH groups in the cellulose I crystalline phase. For the regenerated cellulose sample, the FTIR bands at around 3445 cm^{-1} and the shoulder peak at a lower wavenumber come from the –OH stretching vibration of the inter- and intra-molecular hydrogen-bonded –OH groups of the cellulose II crystalline phase.[86–88] The band at around 2896 cm^{-1} corresponds to the symmetric and asymmetric vibration of CH_2 groups. The bands at 1317–1430 cm^{-1} come from the CH_2 and CH bending. The band at 1165 cm^{-1} is the C–O–C stretching vibration of the β-1,4 glycosidic linkage of cellulose. The bands at 1110 cm^{-1} are attributed to the asymmetric vibration of the glucose ring of cellulose. The bands at 1030 and 1058 cm^{-1}

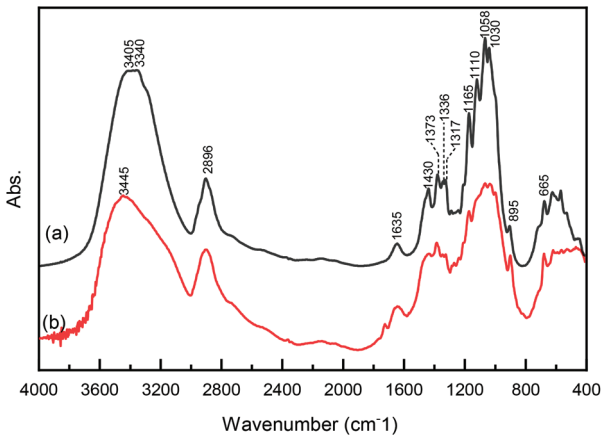


Fig. 7 FTIR spectra of (a) MCC and (b) cellulose sample regenerated from the cellulose/ChOH/Ur solution.

correspond to the C–O(H) strength.[86–88] The bands at 4000–2995, 2900, 1430, 1375, and 900 cm^{-1} are especially sensitive to the state of crystalline and amorphous regions of cellulose.[89] FTIR results indicate that the regenerated cellulose sample has a cellulose II crystalline phase. Fig. 8 shows the typical WAXD curves of MCC and regenerated cellulose from cellulose/ChOH/Ur solution. The WAXD curve of MCC clearly shows the typical diffraction peaks from the crystalline planes of (101), (10$\bar{1}$), and (002) at 2θ of 15.0°, 16.5°, and 22.5°, respectively (Fig. 8a). On the WAXD curve of the regenerated cellulose from cellulose/ChOH/Ur solution, the diffraction peaks at 2θ of 12.1°, 20.0°, and 21.6° correspond to the (101), (10$\bar{1}$), and (002) of the cellulose II crystalline phase. Both FTIR and WAXD results confirm the transition of cellulose I to cellulose II during the dissolution and regeneration of cellulose in the ChOH/Ur system. The dissolution of cotton linter pulp was also checked. The SEM observation indicates that no fibrous shaped cellulose can be found in the regenerated cellulose samples (Fig. S5, ESI†).

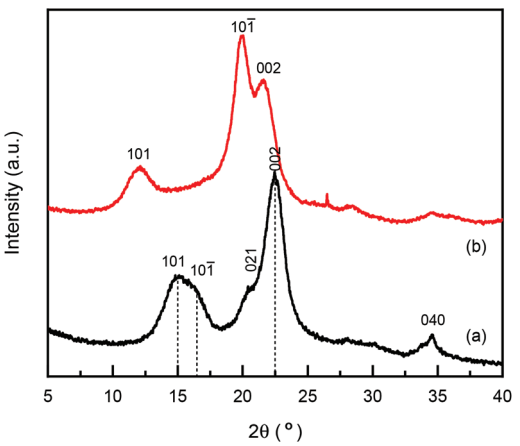


Fig. 8 WAXD curves of (a) MCC and (b) cellulose sample regenerated from ChOH/Ur.

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Physical properties of ChOH/Ur and stability of cellulose in the solutions

The ChOH/Ur system (with *ca.* 11 wt% water) melts at −60.7 to −48.1 °C with the melting peak at −54.8 °C (Fig. 9). Francisco *et al.*[34] also reported the observation of glass transition of many DESs instead of clear melting points by DSC. In the cooling cycle, our result was consistent with theirs. Fig. 10 shows the typical TGA curve and the corresponding derivation curve of the ChOH/Ur DES prepared in this work. The 10% weight loss below 100 °C should be due to the evaporation of water in the system upon heating. Thermal decomposition starts at a temperature above 100 °C. Therefore, the temperature for dissolution of cellulose, the storage of the cellulose solution, and the process for preparing regenerated cellulose materials should be kept below 100 °C. Considering the solubility of cellulose in the ChOH/Ur system as a function of temperature (Fig. 2), the optimum temperature for dissolving cellulose solution in ChOH/Ur (1 : 2) system and subsequent processing for preparing regenerated cellulose materials is around 70 °C.

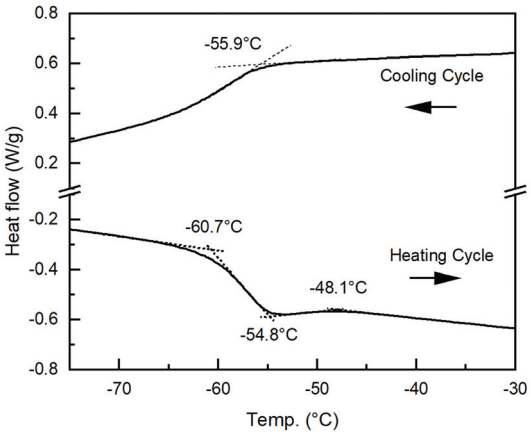


Fig. 9 Typical DSC curves of ChOH/Ur DES with *ca.* 11 wt% water.

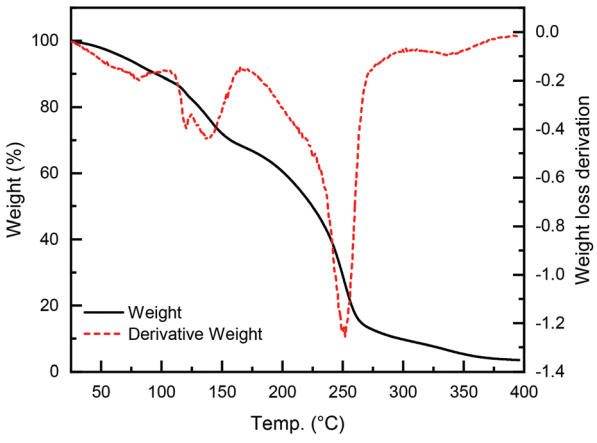


Fig. 10 TGA curves of ChOH/Ur DES with *ca.* 11 wt% water.

Fig. S6 (ESI†) shows the ^{1}H NMR (400 MHz, deuterium oxide as external reference) spectra of the prepared ChOH/Ur DES. The signals at δ = 6.33, 3.84, 3.15, 3.01 ppm correspond to the proton exchange peak (including –NH_2, H_2O, –OH), two adjacent methylene hydrogens, and methyl hydrogens, respectively. The recyclability of the ChOH/Ur DES was also checked. Fig. 11 shows the photos of the as-prepared (origin) and the recycled ChOH/Ur DES. The recycling of ChOH/Ur DES was carried out as follows: the cellulose/DES solution with 5 wt% MCC was prepared and followed by precipitation in a coagulation bath of H_2O/ethanol (70/30, v/v). The filtrate was collected and evaporated. The recycled ChOH/Ur DES was reused for dissolving cellulose. The procedure was repeated many times to evaluate the solubility of cellulose in the recycled ChOH/Ur DES. The first recovery yield is 91%, while the second and the third recovery yields were over 99%. Losses mainly come from the transfer process. After recycling, DES still has a good solubility for cellulose (Fig. S8†). Our results confirm that the ChOH/Ur DES is recyclable.

The stability of cellulose in the cellulose/ChOH/Ur solution was checked. The samples were prepared by regeneration after storage at 40 °C for the desired time. The [η] of cellulose samples was measured in 0.5 M cupriethylenediamine (CED) at 25 °C and the DP of the samples was calculated by eqn (4) (Fig. S7 and Table S2 in ESI†). Fig. 12 shows the DP of cellulose as a function of storage time. The results indicated that the DP of cellulose decreases to around 80% during the dissolution

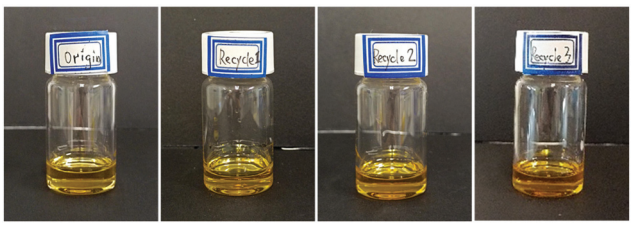


Fig. 11 Photos of as prepared (origin) and recycled ChOH/Ur DES.

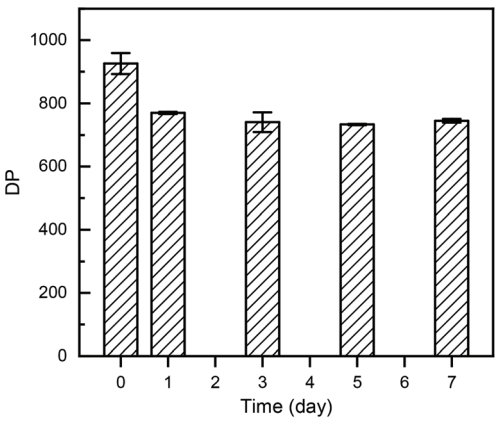


Fig. 12 Histogram of the degrees of polymerization (DP) *versus* dissolution time in ChOH/Ur. The solution was stored at 40 °C.

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process. Storage has no obvious influence on the DP of cellulose. The results confirm that cellulose degradation in the ChOH/Ur DES system is limited, and the ChOH/Ur solvent system is promising for preparing high-performance regenerated cellulose materials.

Conclusions

A systematic investigation has confirmed that almost all the ChCl-based DESs reported in the literature cannot dissolve cellulose. A new ChOH/Ur DES, with ChOH as the HBA and urea as HBD, was found with a sound solubility of cellulose. The ChOH/Ur system with *ca.* 11 wt% water can dissolve at least ~9.5 wt% cellulose with DP of 926 (cotton litter pulp). The maximum value of solubility was achieved at 70 °C within 30 min at the laboratory level. Our results also showed that the ChOH/Ur DES is tolerant to water. ChOH/Ur is a direct solvent system for cellulose, and no derivatizing reaction was found during the dissolution. Kamlet–Taft parameters of the EDSs were measured, and the results indicated that ChOH/Ur has a strong hydrogen bond basicity with the β value of 1.88. The preliminary rheological results suggest that the dynamics of cellulose solution in ChOH/Ur system is similar to that of salt-free polyelectrolyte solutions. The degradation of cellulose during the dissolution and storage is limited. After the cellulose/ChOH/Ur solution was stored at 40 °C for 7 days, the DP of the regenerated cellulose remained about 80% of raw materials. The ChOH/Ur DES is recyclable without losing the solubility of cellulose. The ChOH/Ur DES is a promising solvent platform for preparing regenerated cellulose materials and may also be a media for cellulose chemistry.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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